MUSEUM HEIST

Computer Science I - 2026-I

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Grid-based stealth game integrating → Greedy, Backtracking, Linked List, BST, C++, Python, Pygame, and JSON

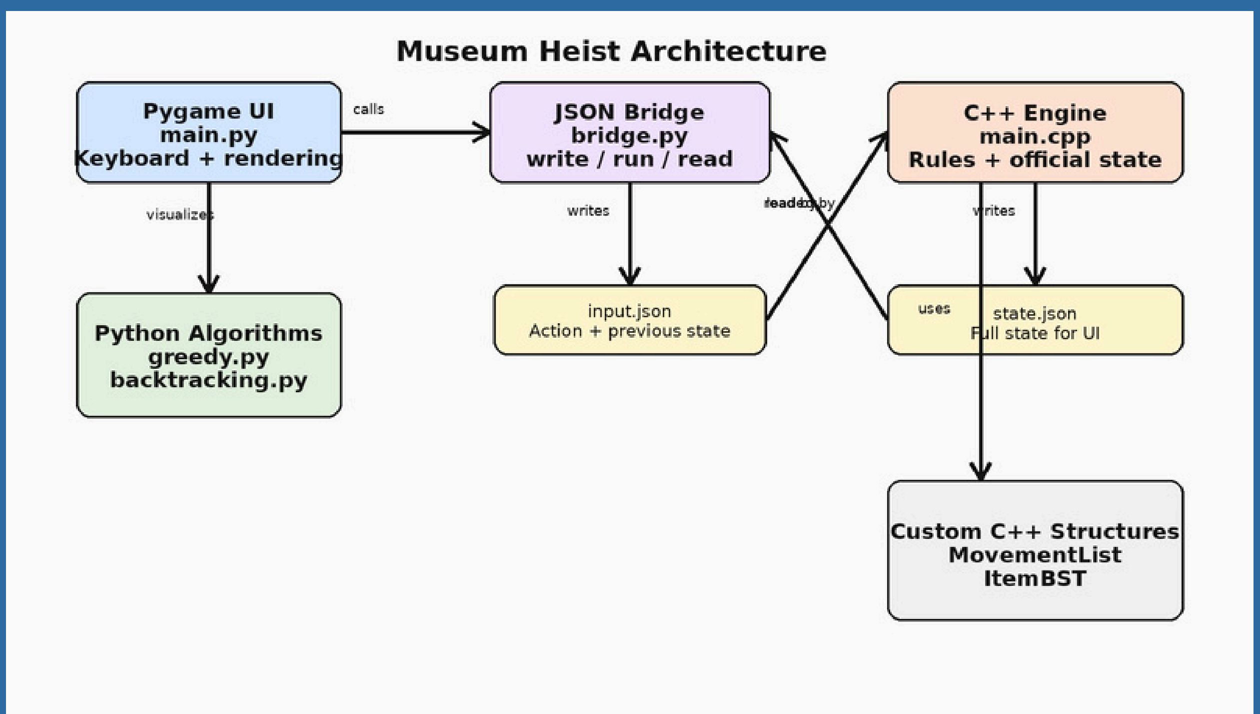
INTRODUCTION

Museum Heist is a playable stealth game built for Computer Science I. The player moves through a museum grid, collects items, avoids surveillance, and reaches the exit before losing by alarm, movement limit, or time limit.The project turns course topics into mechanics: greedy recommends an item, backtracking shows a route, C++ structures store state, and Pygame renders the game.

GOAL

Collect valuable objects, avoid guards, cameras, and vision zones, and then escape through the exit with the best possible final metrics.

ARCHITECTURE



- Pygame UI writes input.json.
- C++ engine processes official state.
- state.json returns data to UI and algorithms.

ALGORITHMS

Greedy - item selection

Chooses one remaining item using immediate local criteria: high value, proximity to guard, and danger status. It is locally optimal, not globally guaranteed.

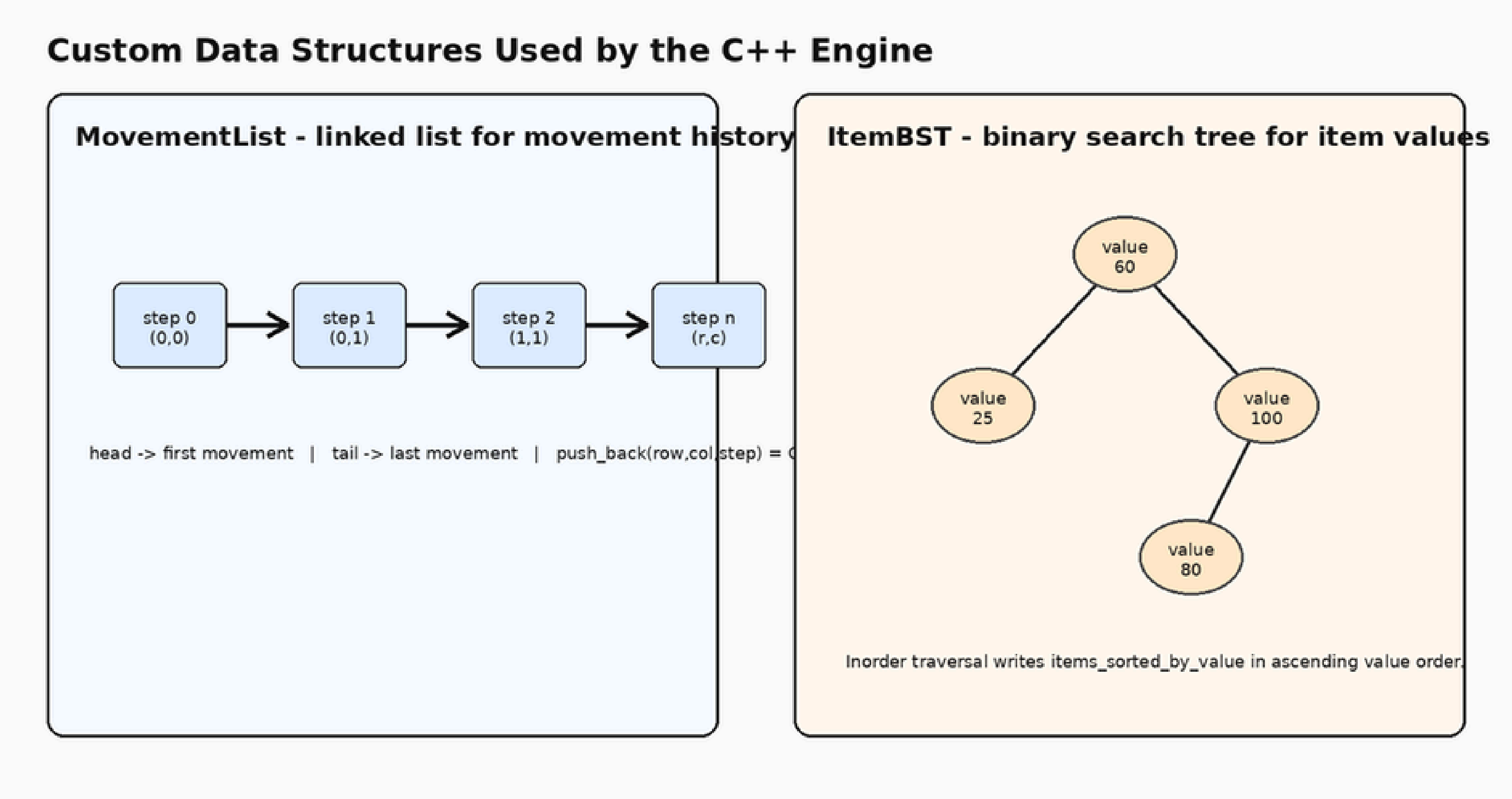
Backtracking - escape route

Uses recursive choose-explore-unchoose logic to find a safe route from the player to the exit while avoiding walls, cameras, guards, and vision zones.

Complexities

Greedy: O(i(g+v)). Backtracking: exponential worst case, bounded in practice by small maps and blocked cells.

DATA STRUCTURE

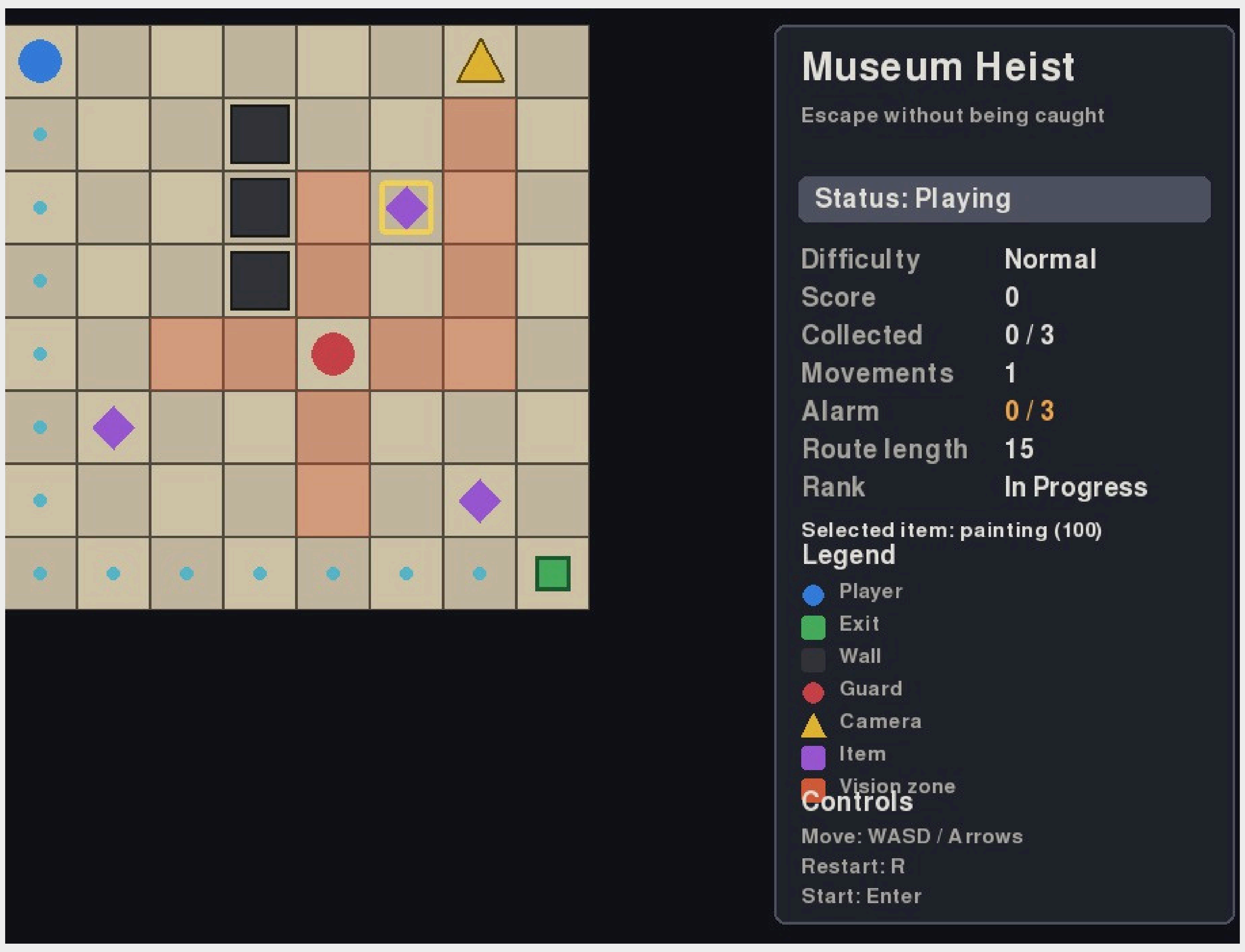


- MovementList: appends valid player positions and writes movement_history.
- ItemBST: orders remaining items by value and writes sorted output.

IMPLEMENTED RESULTS

- Three difficulty levels: Easy, Normal,Hard.
- Keyboard movement with WASD and arrows.
- Camera and guard vision zones with wall blocking.
- Item collection, disappearance, score, and final ranking.
- Loss conditions: vision, alarm, movement limit, and time limit.
- Final screens, tutorial, restart, and main-menu return.

GAMPLEY UI



GAME RULES

- Win: reach the exit safely.
- Lose: enter vision zone, alarm limit, movement limit, or time limit.
- Alarm increases after wall collisions, guard proximity, or risky item collection.
- R restarts the current difficulty. M returns to the main menu.

CONCLUSIONS

Museum Heist demonstrates that **algorithms and data structures can drive a playable system** rather than remain isolated exercises.

The final architecture separates responsibilities clearly: **C++ owns rules and structures**, **Python owns interaction and visualization**, and **JSON connects both layers**.

Future work may include external map files, moving guard patrols, and optimized route search.